

13.2 Circles on the Coordinate Plane

Lesson Introduction

 Benchmarks	MA.912.GR.3.2 Given a mathematical context, use coordinate geometry to classify or justify definitions, properties, and theorems involving circles, triangles, or quadrilaterals. MA.912.GR.3.3 Use coordinate geometry to solve mathematical and real-world geometric problems involving lines, circles, triangles, and quadrilaterals. MA.912.GR.3.4 Use coordinate geometry to solve mathematical and real-world problems on the coordinate plane involving perimeter or area of polygons.		 Learning Targets	- I can use coordinate geometry to determine parts of a circle in a mathematical context. - I can draw a circle using given definitions, properties, or theorems in a mathematical context. - I can use coordinate geometry to solve mathematical problems involving perimeter and area of polygons
 Benchmark Coherence	Building On N/A		Working Toward N/A	
 Essential Questions	- How do you find the lengths of segments of circles on the coordinate plane? - How can you use coordinate geometry to determine parts of a circle? - How do you draw a circle on the coordinate plane given definitions, properties, or theorems in a mathematical context?			
 Lesson Narrative	Lesson 13.2 requires students to apply theorems relating to segments in circles to find the lengths of segments on the coordinate plane. This lesson is an extension of the study of circles in Unit 10. The lesson is also an extension of Lesson 13.1, as students are required to find slopes and write equations of lines in point-slope form and slope-intercept form.			
 Component Summary	13.2.1 Warm-Up (~5 min)	Bell Work (<i>SE p. 288</i>)	13.2.3 Try It (20 min)	Practice on finding segments in circles on the coordinate plane (<i>SE p. 291</i>)
	13.2.2 Exploration (20 min)	Discovering the lengths of segments and parts of circles on the coordinate plane (<i>SE p. 289</i>)	13.2.4 Wrap-Up (~5 min)	Students summarize their learning (<i>SE p. 293</i>)
 Resources	Glossary		Materials	
	Key Terms: - N/A Additional Terms: - Chord - Secant - Slope - Tangent		(Optional) Graphing Utility	
				If planning on using the 13.2.3 Try It as a group activity, prepare the activities for the groups.

 Teacher Tips	Common Misconceptions • Student may struggle with finding coordinates of points when the coordinates are not given. • Students may not substitute coordinates correctly into the formula for point-slope form. • Students may struggle with recalling prior knowledge on secant and tangent lines to help them find the lengths of segments.	Things to Consider • Consider allowing students to use a graphing utility to enhance Lesson 13.2. • Consider reviewing the distance formula and the Pythagorean Theorem, if needed.
	Literacy and Language Routines Compare and Connect Lesson 13.2 gives an opportunity for students to compare the content of the lesson to previous standards on circles and equations of lines. Students will need to apply knowledge and theorems from previous lessons to complete the tasks in 13.2, and to examine various methods for solving problems.	Mathematical Thinking and Reasoning (MTR) Standard MA.K12.MTR.4.1: Discussion and Reflection Students should be engaging in discussions throughout the lesson and should be able to communicate their strategies and methods for solving problems. Students should also be asking questions to their peers and justifying their answers. Lesson 13.2 provides an opportunity for students to engage in discussions in mathematics.
	 Differentiate Instruction	
There are several parts to this lesson that can be differentiated. Consider structuring the Try It Section into a Station Rotation and differentiate by ability level. Question 1 would be a beginner level. Question 2 would be the advanced level, and question 3 would be the intermediate level. Consider assigning students into groups of 2-3 and have two to three groups per question for potentially up to 9 groups working at the same time. Give students a determinate amount of time to complete the problem at their station. Reform groups of 3, where each person in the new group has completed a separate problem and students become the teachers of their problem.		
Lesson Notes:		

13.2.1 Warm-Up: Bell Work

Component Overview: Students solve a question to determine which equation of a line is perpendicular to a given equation. This question relates to lesson 13.1.



13.2.1 Warm-Up: Bell Work

1. Which equation represents a line that is perpendicular to the line $y = -\frac{1}{3}x + 3$?

- A. $3x + y = 4$
- B. $x - 3y = 24$
- C. $y - 3x = -4$
- D. $x + 3y = 3$



Use student understanding to inform your instruction, and to determine where during this and further lessons, remediation may be needed. Make a list of misconceptions and determine which should be addressed whole-group, and which should be addressed in a small group with those who are struggling to be proficient.

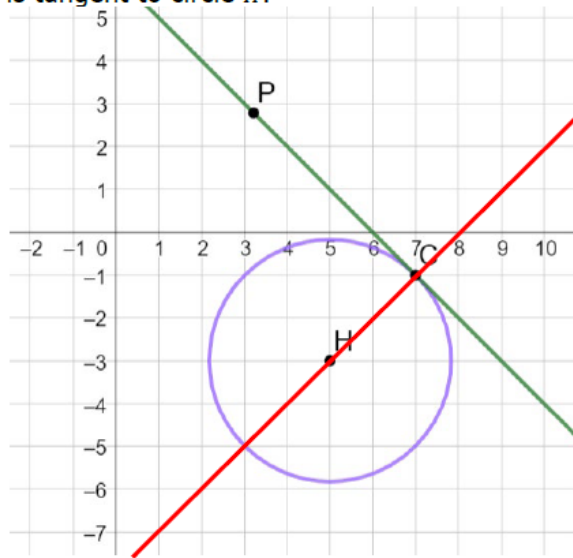
13.2.2 Exploration: Circles on the Coordinate Plane

Component Overview: This Collaboration guides students through finding the lengths of segments and parts of circles on the coordinate plane. Students apply skills from Unit 10, as well as Lesson 13.1. Students will be expected to write equations of lines, find lengths and distances of segments of circles, and find areas and perimeters of triangles.



13.2.2 Exploration: Circles on the Coordinate Plane

- Circle H is shown, where point C is on the circle and line PC is tangent to circle H .



- Use a straightedge to draw a line that goes through points C and H .
See above.
- Discuss with your partner how the slope of tangent line PC and the slope of the radius CH are related.
The slopes of PC and CH are opposite reciprocals.
- What property of circles explains this relationship?
If a line is tangent to a circle, then it is perpendicular to a radius, which ends at the point of tangency.
- Write the equation of line PC .
 $y = -x + 6$



Students should be collaborating throughout the component. Monitor and listen for common student misconceptions in order to identify which students surface those misconceptions, and which students help clarify.



Students may need a refresher on properties of tangent lines. Use the following questions to refresh student memory:

- How do you find the slopes of CH and PC ?
- What do you notice about the slopes?
- What type of angle is formed when a tangent line meets a radius?

13.2.2 Exploration: Circles on the Coordinate Plane (continued)

- e. Write the equation of line CH .

$$y = x - 8$$

- f. Line PC and line CH intersect at point $C(7, -1)$. Use the equations of lines PC and CH to confirm this point.

$$-1 = (-7) + 6$$

$$-1 = 7 - 8$$

2. Circle M has a center at $(-2, 1)$ and a tangent line with the equation $y = 2x - 5$.

- a. Write the equation of the line perpendicular to the tangent line.

$$1 = \frac{-1}{2}(-2) + b; \quad b = 0$$

$$y = \frac{-1}{2}x$$

- b. Find the point where the tangent line touches circle M . Call this point R .

$$\frac{-1}{2}x = 2x - 5; \quad x = 2; \quad y = -1$$

- c. What is the radius of circle M ?

$$\sqrt{(2 + 2)^2 + (-1 - 1)^2} = \sqrt{20} = 2\sqrt{5}$$

- d. Write the equation of circle M .

$$(x + 2)^2 + (y - 1)^2 = 20$$



Consider using a graphing utility for questions 1d-1f, and 2a-2d.



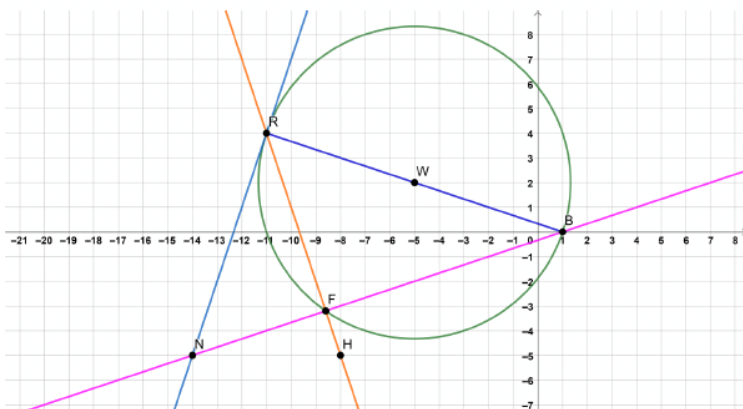
Consider reviewing with students how to write the equation of a line when given the point and the slope. Use MLR 7, and have students Compare and Connect from previous content to the content of the current lesson.



How can you find the length of the radius of circle M ?

13.2.2 Exploration: Circles on the Coordinate Plane (continued)

3. Circle W is shown, with secant lines RH and NB , tangent line NR , and chord RB .



- a. Complete the table.

Line Segment	Length
$\overline{NB}$	$d = \sqrt{(-5 - 0)^2 + (-14 - 1)^2} = 5\sqrt{10}$
$\overline{NR}$	$d = \sqrt{(-5 - 4)^2 + (-14 + 11)^2} = 3\sqrt{10}$

- b. Determine the length of $\overline{NF}$.

$$(NR)^2 = (NF)(NB)$$

$$(3\sqrt{10})^2 = (NF)(5\sqrt{10})$$

$$NF = \frac{18}{\sqrt{10}}$$



To help students get started with the table, ask the following questions:

- How do you find the length of NB ?
- How do you find the length of NR ?
- What strategies can you use to find the lengths of NB and NR ?



Common Misconception

Students may struggle with finding NF because point F is not a whole number, and cannot easily be identified on the coordinate plane. Have students find the equations of lines RH and NB . Set the equations equal to find point F . Then, use the distance formula to find NF .

13.2.2 Exploration: Circles on the Coordinate Plane (continued)

c. Complete the table.

Line	Slope	Equation in Point-Slope Form
$\overleftrightarrow{NR}$	3	$y + 5 = 3(x + 14)$
$\overleftrightarrow{RH}$	-3	$y - 4 = -3(x + 11)$
$\overleftrightarrow{NB}$	$\frac{1}{3}$	$y + 5 = \frac{1}{3}(x + 14)$

d. Rewrite the equation of $\overleftrightarrow{RH}$ in slope-intercept form.
 $y = -3x - 29$

e. Rewrite the equation of $\overleftrightarrow{NB}$ in slope-intercept form.
 $y = \frac{1}{3}x - \frac{1}{3}$

f. Use the slope-intercept forms of $\overleftrightarrow{RH}$ and $\overleftrightarrow{NB}$ to find point F .
 $-3x - 29 = \frac{1}{3}x - \frac{1}{3}; (-8.6, -3.2)$

g. Work with your partner to determine if $\triangle BRN$ is isosceles. Show your work as justification.
It is not isosceles.

$$BR = \sqrt{(-11 - 1)^2 + (4 - 0)^2} = 4\sqrt{10}$$

$$RN = \sqrt{(-5 - 4)^2 + (-14 + 11)^2} = 3\sqrt{10}$$

$$NB = \sqrt{(-5 - 0)^2 + (-14 - 1)^2} = 5\sqrt{10}$$



Consider giving students the formula for Point-Slope Form to assist with question 3c.



Common Misconception

Monitor students' work when substituting coordinates into point-slope form. Continue to monitor students when rewriting the equations of lines RH and NB in slope-intercept form.



Ask the following questions to guide students on question G:

- What mathematical strategy would be used to determine whether triangle BRN is isosceles?
- What is the defining characteristic of an isosceles triangle?

13.2.2 Exploration: Circles on the Coordinate Plane (continued)

- h. What could be used to justify that $\triangle RFB$ is a right triangle?

The slopes can be used to justify that the triangle is a right triangle. If two lines have opposite reciprocal slopes then they are perpendicular and thus intersect at a right angle, creating a right triangle.

The slopes of $\overline{RF}$ ($m = -3$) and $\overline{FB}$ ($m = \frac{1}{3}$) are perpendicular.

- i. What is the area of $\triangle RFB$?

$$RF = \sqrt{(-11 + 8.6)^2 + (4 + 3.2)^2} = \sqrt{57.6}$$

$$NB - NF = FB; 5\sqrt{10} - \frac{18}{\sqrt{10}} = \frac{32}{\sqrt{10}}$$

$$\text{Area} = \frac{1}{2}(RF)(FB); \text{Area} = 38.4 \text{ square units}$$

- j. What is the perimeter of $\triangle RFN$?

$$\text{Perimeter} = RF + FN + NR$$

$$RF + FN + NR = \sqrt{57.6} + \frac{18}{\sqrt{10}} + 3\sqrt{10} \approx 22.768 \text{ units}$$



If necessary, remind students of the formulas for the area and the perimeter of a triangle.

13.2.3 Try It: Circles on the Coordinate Plane

Component Overview: Students will work on various practice problems relating to circles on the coordinate plane. Question 1 has students find a tangent point and the equation of a circle. On question 2, students find distances of segments of circles, then find slopes. Question 3 requires students to find distances, and write the equation of a circle.



13.2.3 Try It: Circles on the Coordinate Plane

1. Circle G has a center at $(9, -2)$ and a tangent line with the equation $3x + 2y = 36$.

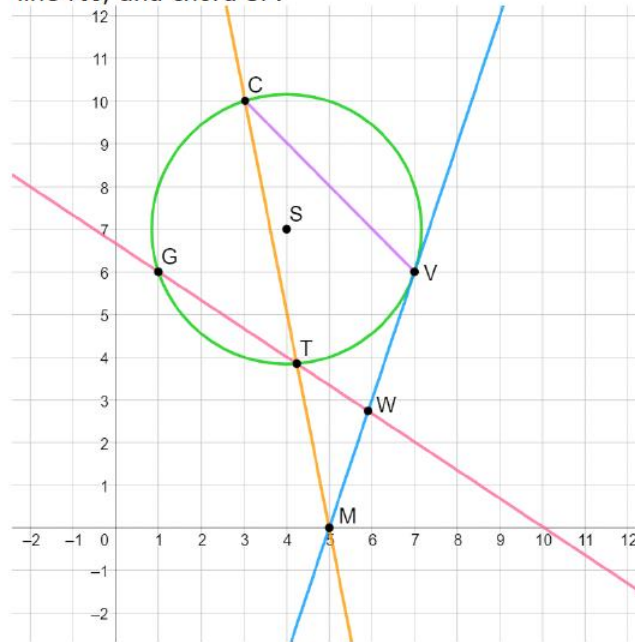
- a. At what point does the tangent line touch circle G ?

$$\frac{-3}{2}x + 18 = \frac{2}{3}x - 8; (12, 0)$$

- b. Write the equation of circle G .

$$(x - 9)^2 + (y + 2)^2 = 13$$

2. Circle S is shown with secant lines CM and GW , tangent line VM , and chord CV .



Consider having students use a graphing utility to construct question 1 for a visual approach. Providing the visual approach can also help refresh students' memories with circle properties.



For a differentiation idea for 13.2.3, see notes in "Differentiate Instruction."



Use the following questions to guide students through question 2:

- What strategy can be used to find the lengths of VM , CV , and CM ?
- Given CM , how can the length of TM be determined?

13.2.3 Try It: Circles on the Coordinate Plane (continued)

a. Complete the table.

Line Segment	Length
$\overline{VM}$	$d = \sqrt{(7 - 5)^2 + (6 - 0)^2} = 2\sqrt{10} \approx 6.325$
$\overline{CV}$	$d = \sqrt{(3 - 7)^2 + (10 - 6)^2} = 4\sqrt{2} \approx 5.657$
$\overline{CM}$	$d = \sqrt{(3 - 5)^2 + (10 - 0)^2} = 2\sqrt{26} \approx 10.198$
$\overline{MT}$	$(2\sqrt{10})^2 = MT(2\sqrt{26})$ $MT \approx 3.922$

b. What is the perimeter of $\triangle CMV$?

$$\text{Perimeter} = CV + VM + MC$$

$$CV + VM + MC \approx 22.18 \text{ units}$$

c. Complete the table.

Line	$\overline{GW}$	$\overline{CM}$
Slope	$-\frac{2}{3}$	-5
Equation in Point-Slope Form	$y - 6 = -\frac{2}{3}(x - 1)$	$y - 0 = -5(x - 5)$
Equation in Slope-Intercept Form	$y = -\frac{2}{3}x + \frac{20}{3}$	$y = -5x + 25$

d. Use the slope-intercept form of both lines to find point T .

$$-\frac{2}{3}x + \frac{20}{3} = -5x + 25$$

$$(4.231, 3.846)$$



Common Misconception

Students may struggle with finding MT since it requires students to use their knowledge on finding segment lengths of secants and tangents of a circle.



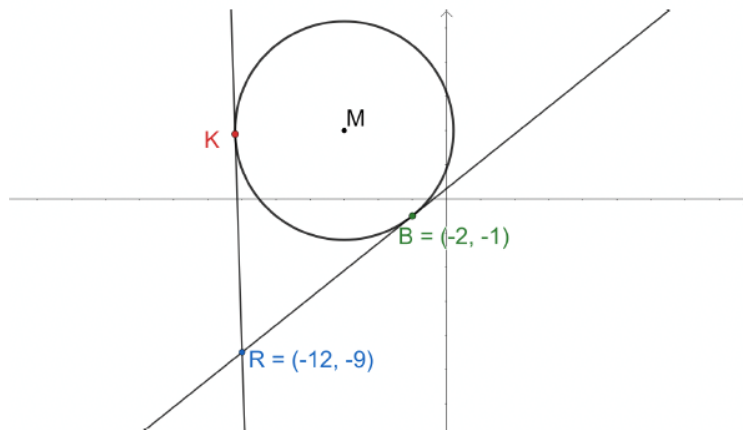
Since point W is not a whole number, consider guiding students through finding the slope of GW .



Students may make errors with fractions when writing the equations of lines for line GW .

13.2.3 Try It: Circles on the Coordinate Plane (continued)

3. Circle M is shown with tangents $\overline{RK}$ and $\overline{RB}$.



- a. What is the length of $\overline{RK}$?

$$RB = RK$$

$$RB = \sqrt{(-2 + 12)^2 + (-1 + 9)^2} = 2\sqrt{41} \approx 12.806$$

- b. The center of the circle is located at $(-6, 4)$. Write the equation of circle M .

$$r = \sqrt{(-6 + 2)^2 + (4 + 1)^2} = \sqrt{41}$$

$$(x + 6)^2 + (y - 4)^2 = 41$$

- c. What type of quadrilateral is $KMBR$? Justify your choice.

Quadrilateral $KMBR$ is a kite because the measure of two pairs of adjacent sides are equal.

$$KM = MB; KR = BR$$

- d. What is the distance between point R and point M ?

$$(2\sqrt{41})^2 + (\sqrt{41})^2 = \sqrt{205}; RM \approx 14.318$$



Consider asking the students the following questions to assist with question 3:

- What segment is RK congruent to in the diagram?
- Why is RB congruent to RK ?
- Given the center of the circle, how can the length of the radius be found to write the equation of the circle?



What characteristics determine that quadrilateral $KMBR$ is a kite?



For a challenge, have students find the coordinates of point K .

13.2.4 Wrap-Up: Cool Down

Component Overview: Students write a short summary of their learning in the lesson. Student responses can be used to identify misconceptions and plan for upcoming instruction.



13.2.4 Wrap-Up: Cool Down

1. Write a note to a classmate who might have missed today's lesson, in your own words, about circles on the coordinate plane.

Answers will vary.



Consider using student responses from prior self-reflections to consider a student's learning progress. Recording this information and pairing it with assessment data provides insightful feedback for student learning progression.